

Classwork (23rd August)

Problem 1

A cloud server is monitored over fixed one-minute intervals. Let X denote the number of request failures in a one-minute interval. The PMF is given by:

X	0	1	2	3
$P(X = x)$	0.6	0.25	0.1	0.05

- (i) Verify PMF.
- (ii) Find $P(X \geq 2)$.
- (iii) Compute $E[X]$, $Var(X)$.
- (iv) Compute Coefficient of Variation (CV).

Problem 2

A bank monitors a portfolio of 4 loans over one year. Let X denote the number of loans that default during this period. The PMF is given by:

$$P(X = 0) = 0.4, P(X = 1) = 0.35, P(X = 2) = 0.2, P(X = 3) = 0.05$$

- (i) Find $E[2X + 3]$, $Var(3X + 2)$.
- (ii) Compute CV.

Problem 3

A photon detector measures the number of photons arriving per second. Let X denote the number of photons detected in one second. The PMF is given by:

$$P(X = 1) = 0.2, P(X = 2) = 0.5, P(X = 3) = 0.3$$

- (i) Find expectation and variance.
- (ii) Compute CV.

2. PDF (Continuous Random Variables)

A PDF satisfies:

$$f(x) \geq 0, \int f(x) dx = 1$$

Problem 1

A cloud server's response time is measured in seconds. Let X denote the response time for a request. The probability density function of X is:

$$f(x) = 2e^{-2x}, x \geq 0$$

- (i) Find $P(X < 1)$.
- (ii) Compute $E[X]$, $Var(X)$, CV.

Problem 2

In an electronic communication system, let X represent the normalized amplitude of noise (scaled between 0 and 1). The probability density function of X is:

$$f(x) = 3x^2, 0 \leq x \leq 1$$

- (i) Verify PDF.
- (ii) Find $E[5X + 1]$, $Var(2X + 5)$, CV.

Problem 3

In an economic model of income distribution, let X represent income (in normalized units), where $X \geq 1$. The probability density function of X is

$$f(x) = kx^{-2}, x \geq 1$$

- (i) Find k .
- (ii) Compute $E[X + 5]$, $Var(X - 5)$, CV.

Problem 4

In a physical system, let X represent the normalized energy of a particle, where $0 \leq X \leq 1$. The probability density function of X is:

$$f(x) = 2x, 0 \leq x \leq 1$$

- (i) Compute expectation and variance.
- (ii) Compute CV.

3. CDF (Cumulative Distribution Function)

$$F(x) = P(X \leq x)$$

Problem 1

In a reliability study, let X represent the lifetime (in hours) of a device. The cumulative distribution function of X is:

$$F(x) = 1 - e^{-x}, x \geq 0$$

- (i) Compute $P(X \leq 2)$.
- (ii) Find PDF.
- (iii) Compute $E[X]$, $Var(X)$, CV.

Problem 2

Let X represent the normalized lifetime of a battery, where $0 \leq X \leq 1$. The cumulative distribution function of X is:

$$F(x) = x^2, 0 \leq x \leq 1$$

- (i) Find PDF.
- (ii) Compute $E[X]$, $Var(X)$, CV.

Problem 3

In an economic study of income inequality, let X represent income (normalized such that $X \geq 1$). The cumulative distribution function of X is:

$$F(x) = 1 - \frac{1}{x}, x \geq 1$$

- (i) Find PDF.
- (ii) Compute $P(1.5 \leq X \leq 3.5)$.
- (iii) Compute expectation and variance.

Problem 4

Let X represent the lifetime (in hours) of a system component. The cumulative distribution function of X is:

$$F(x) = 1 - e^{-\lambda x}, x \geq 0$$

- (i) Find PDF.
- (ii) Compute mean and variance.
- (iii) Compute $P(0.25 \leq X \leq 1.5)$ and $P(X \geq 3.5)$.

Problem 5

In a quality control process, let X represent the number of defective items in a batch. The cumulative distribution function of X is:

$$F(1) = 0.2, F(2) = 0.6, F(3) = 1$$

- (i) Derive PMF.
- (ii) Compute $E[X]$, $Var(X)$, CV.